Assignment 8

Self-organizing maps: a controlled comparison of neighborhood functions
Machine Learning in Finance | FRE-GY.7773.C | Unsupervised learning and clustering

Experimental design

I implement the assigned SOM experiment on the 150-observation, four-feature Iris dataset. Each feature is MinMax scaled to [0,1]. The 8x8 map is rectangular, uses Euclidean BMU distance and PCA initialization, and performs 5,000 random-sample updates. Learning rate starts at 0.5 and decays linearly toward zero; sigma starts at 1.5 and decays linearly toward one. The primary random seed is 123.

Only the neighborhood function changes: **Gaussian, Bubble, Mexican hat and Triangle**. All runs share the initial prototypes and random training-sample order. For the Fig. 3-style plots, the same seeded display jitter is applied to every model, and class indices 0,1,2 are correctly mapped to species. Labels are used only for visualization and post-training diagnostics, never for learning.

Quantization error (QE) is the **mean Euclidean distance** from each normalized observation to its best matching prototype: $QE=(1/N)\ \text{sum}\ ||x_i-w_{BMU(i)}||$. I cross-check MiniSom's QE with this direct definition. This is a descriptive unsupervised fit, not held-out prediction accuracy.

Results for the common primary seed

Neighborhood	QE	Topographic error	Occupied cells	Mixed cells	Cell purity (%)
Gaussian	0.094291	0.0133	51	2	98.67
Bubble	0.091690	0.0333	52	4	97.33
Mexican Hat	0.251714	0.0000	2	2	66.67
Triangle	0.054756	0.1200	57	4	97.33

Table 1. Seed 123. A mixed cell contains more than one species. Cell purity is the fraction assigned to each occupied cell's majority species. Topographic error follows MiniSom's rectangular-grid convention, including diagonal adjacency.

Triangle has the smallest QE (0.054756), followed by Bubble (0.091690) and Gaussian (0.094291). Triangle uses 57 cells and gives a finer quantization, but its topographic error is larger than Gaussian's. Thus low quantization error need not mean the best preservation of neighborhood relationships.

Robustness across five matched seeds

Neighborhood	QE mean	QE sample SD	QE min-max
Gaussian	0.093360	0.000817	0.092402 - 0.094291
Bubble	0.092040	0.000600	0.091495 - 0.093046
Mexican Hat	0.252897	0.002685	0.249008 - 0.255528
Triangle	0.055121	0.000336	0.054756 - 0.055667

Table 2. Five training-order seeds, 123-127, with the same paired comparison in every seed. The QE ranking is unchanged, so the primary result is not an isolated seed effect.

A failure case is part of the comparison

Mexican hat has a repulsive outer ring. Under these **fixed** learning-rate and radius settings, many unused prototypes diverge: the primary run reaches a maximum absolute weight of 4.38×10^{192}, while only two cells receive data. Its QE is 0.251714 and purity is 66.67%. Its zero topographic error is a degenerate two-prototype outcome, not evidence of a good map. The same failure occurs in all five seeds. I retain it rather than retuning this function and changing the controlled comparison.

Sigma=1.5 is kept for Bubble/Triangle despite MiniSom's noninteger-radius warning. No trained weights are clipped. BMUs are verified with overflow-safe hypot distances. Common rescaling of prototypes cancels during U-matrix normalization and prevents display overflow.

Comparison of data points within cells (assigned Fig. 3)

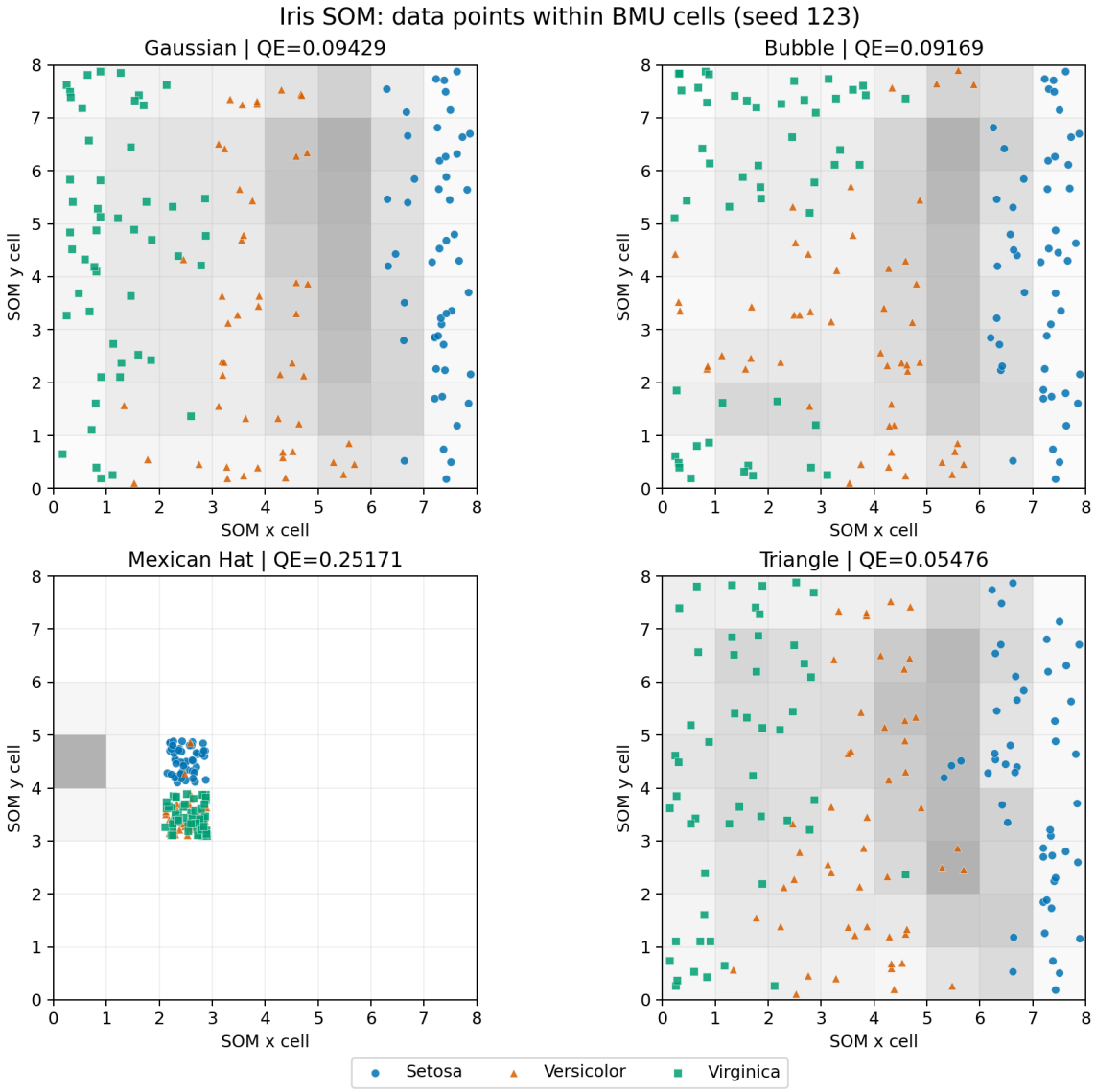


Figure 1. All 150 observations in their BMU cells, with identical within-cell jitter. The light-to-dark background is each map's U-matrix normalized to [0,1]. Darker cells indicate larger summed distances to neighboring prototypes; absolute U-matrix magnitudes are not compared across panels.

Gaussian: Setosa forms a separate strip at the right. Versicolor and Virginica occupy neighboring central/left regions with limited mixing. Its smooth neighborhood gives the lowest nondegenerate topographic error, at the cost of coarser quantization.

Bubble: Setosa is still separated, but the other two species are more scattered and four occupied cells are mixed. The flat neighborhood updates all included cells equally and gives slightly smaller QE than Gaussian under this fixed radius schedule.

Mexican hat: Data collapse into two adjacent cells. One mostly represents Setosa, while the other merges Versicolor and Virginica. The diverging unused prototypes dominate the U-matrix scale. This is an unstable result, not successful cluster separation.

Triangle: The data occupy more cells, Setosa remains largely distinct, and Versicolor/Virginica retain a transition region with some mixed cells. Its compact neighborhood gives the best prototype fit here, while the higher topographic error warns against selecting a map from QE alone.

Conclusion. For the specified tutorial setup, Triangle is the best choice if the objective is QE; Gaussian is preferable when local topology preservation is more important. Bubble is a reasonable intermediate result. Mexican hat would require separate learning-rate/radius tuning before practical use. These conclusions are conditional on this dataset and the shared hyperparameters.

References and reproduction. DataCamp, Self-Organizing Maps: An Intuitive Guide with Python Examples; MiniSom source; UCI Iris dataset. Tested with Python 3.12.1, MiniSom 2.3.6, NumPy 2.1.1, scikit-learn 1.9.1 and Matplotlib 3.9.2. Code, individual plots, five-seed QE figure, prototype weights and full CSV/JSON results accompany the report.